

Simulation of Vertical Filling Processes of Granular Foods for typical Retail Amounts

Stefan Kirsch, André Philipp

stefan.kirsch2@bosch.com, andre.philipp@bosch.com

Bosch Packaging Technology B.V.

Group Food Vertical

Department for Engineering and Development (END)

Abstract

Vertical baggers play an important role in the packaging of food granulates such as bite-sized snacks, candy and nuts. Increasing demands regarding productivity and throughput pressure manufacturers to employ increasingly sophisticated engineering tools over conventional design through trial and error and experience. The Discrete Element Method (DEM) is a powerful tool for prediction of bulk behavior, however randomness in the packaging process impairs reproducibility of simulations, especially when working in the typical range of particle counts less than 1000.

This study looks into statistic methods to determine under which conditions adequate reliability of simulation results can be achieved. Both the influence of the goal variables used to describe the process and the influence of the particle count are discussed. The examples given can be adapted to similar scenarios and lay the foundation for future simulation assisted design of vertical filling processes.

1 Introduction

1.1 VFFS Machines

The vertical form fill sealing process (VFFS) can be found in a variety of industrial applications for packaging of particulate materials. Since the process is capable of high throughput and able to meet high hygiene standards, it is widely used for packaging of granular foods such as bite-sized snacks, candy and nuts. The setup of the process is shown in Figure 1. The top part is responsible for dosing the bulk material. It consists of a weighing unit (not shown), that feeds a collection bin where the particles of one portion are caught and their vertical speed is equalized to (near) zero. The bottom of the tray then opens up, releasing the particle portion. The free falling particles are caught by a funnel and are fed into the vertical, so called forming tube. While the bulk good travels

downwards, a plane film is shaped into a tube and transported down along the forming tube. The film tube is sealed at the bottom, so that the bag can catch the particle portion as it arrives. The bag is then sealed on the top, cut from the film strain and ejected. Sealing is performed by hot plates, known as sealing jaws, which press the film tube together, partially melting the plastic and rendering the bag airtight. In the simplest case, a pillow shaped bag is produced this way.

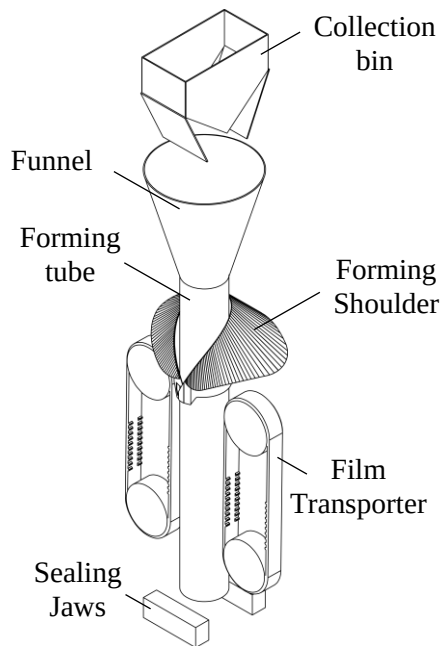


Figure 1: VFFS Process

Because high bag output rates are economically desirable, VFFS manufacturers aim to reduce the safety time buffers for each of the process steps to their respective minimum. One important factor for overall production speed is the time buffer between the sealing events of the bottom and the top part of the bag. This time buffer cannot be arbitrarily short, because the bulk material needs a certain time to fall through the tube and pass the sealing jaws. If it is chosen too short, a particle might get caught between the sealing jaws in some cycles. Each time this occurs, the bag is damaged and must be discarded. Also, damage to the sealing jaws may occur, which may require maintenance and a temporary production stop. A crucial part of VFFS machine configuration is therefore adjusting the fill time buffer just so that process safety is retained, while the desired throughput is maintained. Today this is mostly achieved by extensive testing and combined with the process engineers' experience in choosing appropriate machine parts. Depending on portion and bag size, throughputs of more than 200 bags per minute can be achieved this way. However physical testing is limited to a hand full of designs due to time and cost constraints, so

iterative fine adjustment is rarely possible. Thus generous time buffers between the drop events must be chosen to ensure process safety, which in turn affects bag production rate.

1.2 Granular Systems

Randomness in Granular Dynamics Every contact between particles or between a particle and the boundary can be seen as an event, That will result in an outcome different from the one than if that event had not happened. The initial conditions at which an observation is started, i.e. the particles' position, orientation, translational and rotary speeds play a major role in determining which precise events will occur as the process plays out. Even slight changes in the initial state can entirely change the nature of a given collision event between two particles, or even if it will happen at all. Thus, slight changes in initial conditions can result in dramatically different outcomes. Granulate processes are therefore considered to be chaotic systems [1, 2].

Whether this instability in outcome is critical for process design depends on the goal variables most suitable to assess the respective process' performance. If they depend on the properties of many particles, the individual variations between particles should cancel out to a certain degree and it should be expected that the goal variables will be rather robust to small random changes. On the contrary, if the goal variables highly depend on the history of few or even a single particle, the process should be quite sensitive to initial randomness. Furthermore, processes involving fewer particles tend to, generally speaking, be more sensitive to initial randomness, because there are less particles that can influence the goal variables in the first place.

Practical Implications of Randomness In the case of vertical packaging of granular goods, the precise initial conditions of every cycle, i.e. the individual particles' trajectories and thus their position in the instant of sealing are unknowable. Therefore, dynamically adjusting the fill time buffer depending on how much time is necessary at a particular cycle is not an option. Instead, a for the most part constant estimation for the fill duration in a hypothetical worst case, must be assumed. This however can only be an informed guess, because to completely rule out particles getting caught between the sealing jaws, it would require testing every possible initial condition of the filling cycle, which is not practical.

The effects of randomness in granular systems carry over to simulations, rendering the outcome equally random. Additionally, it is often not possible to exactly reproduce the initial conditions of an experiment to a simulation due to impractical time effort. In order to accurately reproduce an

experiment in a simulation, the randomness seen in real life should instead be recreated as faithfully as possible.

Randomness is especially problematic in numerical optimization, where the trial and error method is performed virtually, so a larger amount of candidate designs can be tested. When simulation outcomes do not only vary due to changes to the design, but also randomly, the optimizer might not be able to determine the gradients correctly and might be sent into a wrong direction. This may slow down convergence and/or reduce the quality of the end result. Several studies suggest methods to increase optimizer robustness [3, 4], but uncertainty remains a challenge. [5]

If on the other hand, the real life randomness is not taken into account in the simulation, one would run the risk of optimizing only the process playing out at particular initial condition, which could vary significantly from the true optimum for all possible initial states.

Numerical Noise Numerical errors are an omnipresent phenomenon in computer calculations. They originate from the inherent limited precision of floating point variables and the consequent rounding during calculations. Usually, the standard double precision is sufficient to produce results with the desired accuracy, so most users will not be confronted with accuracy issues [6, 7].

However, simulations of dynamic systems in which consecutive states are calculated based on previous ones, are generally prone to numerical noise. Every state is affected by the rounding errors of the previous calculation steps, so errors may accumulate over time. For mostly linear systems, again, there should not be any issue, because the error in the final result will be around the same order of magnitude as the error in every individual step. In highly nonlinear systems however (with granular systems being a part of), the effect of the accumulated errors can be substantial [8].

Numerical errors depend on the processor type. This means that if the calculations are performed on one particular processor core, the numerical error will be identical in any number of reruns. In parallel computing however, the calculations are spread out over several cores, which randomly reassigns their order. Due to the fact that associativity and distributivity do not apply for noisy variables, the error will then be different in every simulation run. [6, 7]

Interestingly enough, numerical errors are rarely discussed in studies on particle simulation. One reason could be, that numerical errors, just as the influence of randomness in initial conditions, play a smaller role when more particles are involved in determining the goal variables of a process. Such scenarios are indeed more common in literature [9].

1.3 Discrete Element Method

Particle contacts, even for simple geometries, are very complex in their details [10, 11]. As a consequence, computational cost remains a limiting factor in bulk material simulation up to this day. For the simulation of practical particle quantities, it is necessary to find viable approximations for the contact physics to reduce the computational time to practical levels. In the Discrete Element Method (DEM), this is achieved by discretizing the time domain into finite time steps and by approximating the forces generated during contact with virtual spring and/or virtual dashpot behavior. The resulting equations can then be solved relatively quickly, allowing the simulation of the interactions of many particles at once.

The DEM was originally proposed by Cundall and Strack in 1979 and has since then been continuously refined with respect to both accuracy and computational feasibility [12, 13, 14]. With increasing hardware capabilities, the method went over the years from being capable of simulating a hand full of particles, to today assisting in designing intricate processes with millions of particles involved [15, 16].

There is no shortage in DEM codes on the market, both open source and commercial. The different implementations vary, among others, in the way the contact physics are implemented and how non-spherical particle shapes are represented. For this study a commercial code using the partially latching spring model as described in [17] is used. The repulsive forces during particle collision are calculated from the Hookeian spring law. The model was first proposed by Walton and Braun [18] and is also known as the linear hysteresis model [19].

As two particles begin to overlap, the displacement δ is counted up. The repulsive force in normal direction F_n is calculated according to Equation 1:

$$F_n = k_1 \delta \quad \text{during loading} \quad (1)$$

k_1 is the virtual spring's stiffness during loading. To account for the lossy nature of the collision, the stiffness k_2 is greater on the rebound. The relationship between k_1 and k_2 is expressed via the coefficient of restitution ϵ .

$$\begin{aligned} F_n &= k_2 (\delta - \delta_0) \\ &= \frac{k_1}{\epsilon^2} (\delta - \delta_0) \end{aligned} \quad \text{during unloading} \quad (2)$$

Due to plastic deformation, the particles will separate at a remaining displacement δ_0 . Here, the overlap vanishes and the repulsive force becomes zero. A plot of repulsive normal force over displacement for $\epsilon = 0.4$ is shown in Figure 2.

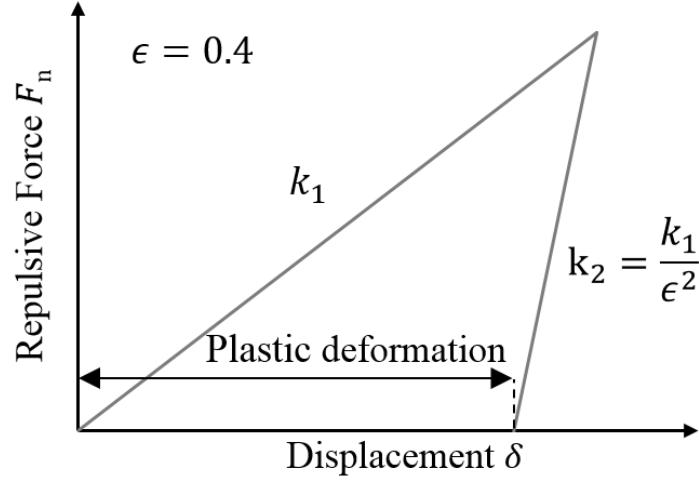


Figure 2: Force versus overlap for partially latching spring contact model (also known as linear hysteresis model)

For the tangential force at the n -th time step of the collision, the following relationship is used:

$$\begin{aligned} F_t^n &= F_t^{n-1} + k\Delta\delta & \text{if } F_t \leq \mu F_n \\ F_t^n &= \mu F_n & \text{if } F_t \geq \mu F_n \end{aligned} \quad (3)$$

μ is the friction coefficient, which is lower when sliding occurs than while sticking. To account for both cases, static and dynamic friction coefficients are defined in the model.

1.4 Filling of typical Retail Quantities

Typical consumer-sized bags of candy, nuts and other bite-sized snacks usually contain well below a kilogram of produce. Expressed in particle count, we can expect to find mostly less than 1000 particles in one bag (ignoring secondary particles such as material dust and broken off chunks).

Process Design The criterion for process safety of the filling process is that in each cycle, no particle gets caught between the sealing jaws as they close. Thus, the time buffer between sealing events necessary to ensure complete filling depends on the total time that the granulate portion

needs in order to fall past the jaws. This value is identical to the statistical range of particle residence times within a portion, so we shall call it the portion range τ_{rg} . It depends on exactly two particles, namely the first one in the portion that passes the jaws and the last one before sealing is performed.

$$\tau_{rg} = t_{\text{res,lp}} - t_{\text{res,fp}} \quad (4)$$

$t_{\text{res,fp}}$ is the residence time of the first, and $t_{\text{res,lp}}$ is the residence time of the last particle of the portion passing the sealing jaws. The residence time is the time between the opening of the collection bin and the time at which the particle leaves the space between the sealing jaws, in which it could potentially get caught.

Stability of Results Since τ_{rg} depends on few (namely two) particles, it can be expected to be quite sensitive to both physical and numerical randomness. This means that during optimization, there is a risk of discarding an actually good candidate solution, that performed poorly just because of “bad luck”. On the other hand, bad candidate solutions can seemingly perform good if the randomness plays out favorably. When performing simulations of the process, we are thus facing the issue, that even if the physical behavior of the bulk good is accurately reproduced, not much useful information is provided to assist in process design.

One way to avoid this issue is to conduct several simulation runs of the same scenario and reproduce the randomness one would expect from real life. For a human engineer it then becomes easier to distinguish, whether a particular result is just an outlier or if it follows a real trend. However, since there is often a practical need for optimization tasks to be done by a computer (because like that, many different designs can be tested automatically), one is often required to assign a numeric value for the quality of a particular design. One way would be to average the portion range τ_{rg} between the several simulation runs. This would increase the chance of a good result actually stemming from a favorable design, rather than just from good luck (and vice versa). However one must determine, how often the simulation should be rerun, so the uncertainty in the result is reduced to an acceptable level, while reasonable runtime is maintained. If runtime is critical, more robust goal variables, i.e. ones that are less prone to randomness, may be used instead. In the following we will look at the influence of the choice of goal variables, particle count and number of averaged reruns on result stability.

2 Material and Methods

2.1 Drop setup

In order to reproduce typical process conditions, including randomness in initial states, a representative trial is used. The precise dimensions, shape and operation characteristics of a real packaging machine are not considered relevant for the problem of this study, so we build a simplified setup that allows quick and reproducible trials with different bulk quantities. Transparent PMMA is used for all parts that come in contact with the bulk material, so the trials can be evaluated visually.

The setup parts are mounted between a back and a front plate with side plates as boundaries for each section, giving them a rectangular base (Figure 3). The sample bin is 100 mm deep and can hold up to one kilogram of bulk material. Two rotatable flaps, that can be opened with a servo motor, are mounted on its bottom. The bin can be fully opened in around 100 ms.

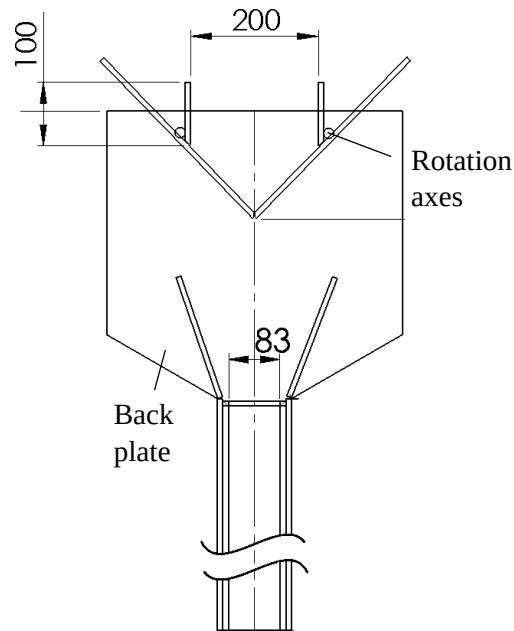


Figure 3: Drop Setup (measures in mm)

The example food is bite-sized chocolate candy with a porous cookie core. As typical retail fill masses, 250, 500 and 700 grams are used respectively.

In order to generate a defined initial state that can be reproduced in the simulations, the fill line is levelled and horizontally aligned manually before every drop. A computer model of the drop setup is prepared and imported into the DEM environment.

2.2 Simulation

Goal Variables As discussed above, we expect the portion range τ_{rg} to be quite sensitive to randomness. Frank et al. faced the same issue and stated that this might lead to unstable results when only using one single simulation run. They therefore used an alternative formulation, which they obtained by recording the residence times of all particles within a single portion and then calculating the standard deviation between the particles' residence times t_{res} . [7] In the following, we shall name this value τ_{sd} . Residence time is not normally distributed, so τ_{sd} has no direct physical meaning. A drawback of this goal variable is, that it is not directly suitable for the prediction of successful machine operation. This is because it is quite insensitive to “outliers”, i.e. particles that separate from the main bunch. This could lead to a good simulation results (i.e. a low value of τ_{sd}), even though, a single delayed particle is getting caught between the sealing jaws. However, it allows a good estimate of how compact the particle portion falls.

In this study, both the portion range τ_{rg} as well as the potentially more robust, but less practical standard deviation τ_{sd} , are considered. Additionally we will consider the average residence time $\overline{t_{res}}$ of particles within the individual granulate portion. An increase of the latter would mean that, all other things being equal, the whole portion is delayed, leaving the tube at a slightly later time, while maintaining its shape and compactness.

Statistical Considerations In order to assess how sensitive the simulation results will be to randomness, we first need to define meaningful statistical parameters, that can be applied to different goal variables. Generally speaking, the more simulation runs are conducted and their results averaged, the closer the average result $\bar{x}_{set,1}$ of the set of runs will be to the statistical expected value $\hat{x}$. The standard deviation $\sigma_{run,n}$ of the result between n runs then allows to assess the variation introduced by randomness.

If we rerun the simulation again n times we obtain another averaged result $\bar{x}_{set,2}$, which should slightly differ from the first result $\bar{x}_{set,1}$. This is due to the fact, that we would need an infinite amount of simulation runs for the averaged result $\bar{x}_{set,i}$ to become the actual expected value $\hat{x}$.

To assess how close the estimate is to the expected value $\hat{x}$, we can calculate the standard deviation $\sigma_{set,m}$ between the averaged results $\bar{x}_{set,1}, \bar{x}_{set,2}, \dots, \bar{x}_{set,\frac{n}{m}}$ of several sets of m runs. The described method is shown schematically in Figure 4.

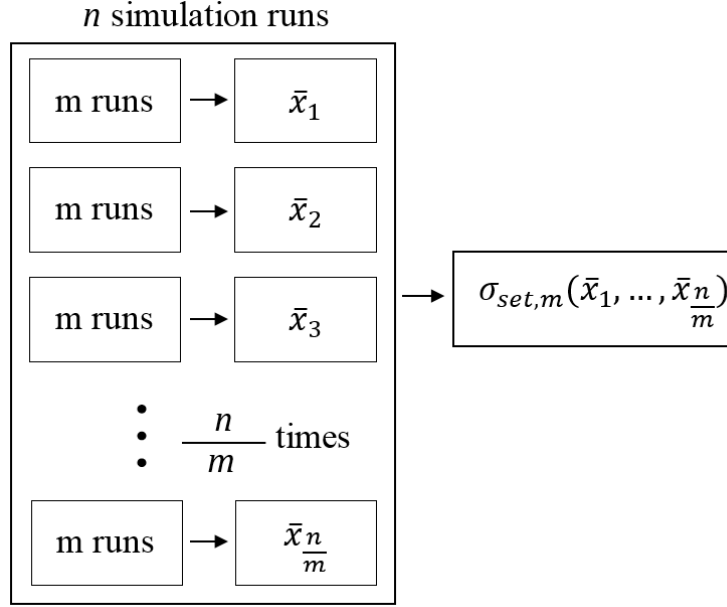


Figure 4: Scheme to divide n runs into m sets and to calculate the mean goal variables $\bar{x}_i$ per set and the standard deviation $\sigma_{set,m}$ between sets.

$\sigma_{set,m}$ has practical meaning, indicating how reproducible (and therefore reliable) a result obtained from averaging m simulation runs is. For example, if $\sigma_{set,m}$ is quite high, performing more reruns should be considered. This will then further average out the randomness and improve the estimate of the expected value $\hat{x}$.

Simulation Model The discrete element method is capable of handling various particle shapes, including spheres, cylinders and arbitrary polyhedrons. While the contact laws for spheres are fairly simple and computationally cheap, the cost increases significantly when using more complex shapes [20]. The chocolate candy pieces are nearly round, so spheres are used in this study. For the sphere diameter, the average sieve diameter of the candy pieces is used. Size distribution is not considered in the model. To make up for the fact that the real particles are not perfectly smooth, we introduce a coefficient for non-physical rolling resistance, which avoids unrealistically strong rolling behavior [21].

The DEM model requires parameters representing the particle material's density, Young's modulus, static and dynamic friction coefficients and the coefficient of restitution. The same set of

parameters is necessary for the boundary material (in this case PMMA). A common issue with DEM models is that material parameters are not available in literature and are difficult to determine analytically [9, 22]. For the relative influence of initial randomness and numerical noise, the precise model parameters are however considered not important. For the purpose of this study, we therefore mainly work with approximate material parameters from similar materials than the chocolate candy. However, to make sure the model is physically viable, we compare the simulation results to high speed-footage of the real-life experiments.

The candy's density can be easily determined in a water displacement test. For static friction between PMMA and the particles, we conduct a simple experiment on an inclined plane. The coefficient of friction can be calculated according to Equation (5) from the angle α at which sliding initiates.

$$\mu_{S,P-W} = \tan \alpha \quad (5)$$

Simulations with a very low coefficient of sliding friction $\mu_{D,P-P}$ are found to sometimes produce unphysical results. This is avoided by setting the value to 0.1. The remaining parameters are obtained from literature. The list of parameters used for the chocolate candy are shown in Table 1.

The randomness in the initial state before the drop is achieved by generating the particles at random positions vertically above the sample bin and then letting them fall through gravity and settle into the sample bin. The fill line is then horizontally flattened.

Runtime In order to isolate the two sources of random variation from each other (numerical noise and randomness in initial conditions), different simulation settings are used to exclude one of them at a time. To generate a random initial state, the above described filling step must be simulated on every run, which increases runtime. On the other hand, to produce constant numerical errors, the scenario is ran on a single core, which reduces computational performance, again increasing runtime. The different scenarios to be simulated are shown with their respective runtime in Table 2.

Several simulation runs per scenario must be conducted to allow for statistical analysis of randomness. For each scenario, the simulations are repeated 300 times on an Intel® Core™ i7-6820HQ processor with four cores. Depending on the scenario, one session of 300 runs took between 4.5 and 32.4 real time hours.

Parameter	Symbol, Unit	Value	Source
Diameter	d [mm]	12.45	Measured
Density	ρ [kg/m ³]	1044	Measured
Young's modulus	E [N/m ²]	10 ⁸	[23]
Restitution	ϵ_{P-P} [-]	0.1	[23]
	ϵ_{P-W} [-]	0.1	[23]
	$\mu_{S,P-P}$ [-]	0.8	[23]
Static friction	$\mu_{S,P-W}$ [-]	0.45	Measured
	$\mu_{D,P-P}$ [-]	0.1	[24], increased for stability
	$\mu_{D,P-W}$ [-]	0.1	[24]
Dynamic friction			
Rolling resistance	RR [-]	0.1	Estimated

Table 1: DEM Model parameters

Initial state	Cores	Runtime in minutes		
		250 g	500 g	700 g
Random	1	2.35	4.38	6.48
Constant	4	0.90	1.12	1.86
Random	4	1.33	2.25	4.33

Table 2: Simulation runtimes per run on Intel® Core™ i7-6820HQ

3 Results

3.1 Comparing Simulation and Experiment

A comparison between experiments and the respective simulations for 250 and 500 g of bulk food respectively are shown in Figure 5 and Figure 6. We observe a good match regarding position and shape of the bulk. This lets us conclude, that with the chosen parameters, the DEM model allows good representation of the real life behavior. However, we find that positions of individual

particles differ between simulation and experiment to a notable extend. Apart from possible model shortcomings, this can be explained by the overall physical randomness of the process, which we do not expect to allow prediction of individual particles' positions.

3.2 Influence of Particle Count

In postprocessing the above introduced goal variables are calculated from the 300 simulation runs per scenario: the portion range τ_{rg} , the average residence time $\overline{t_{res}}$, and its standard deviation τ_{sd} . In Figure 7, the normalized goal variables are plotted against sample mass. We find that all three goal variables increase linearly with sample mass (i.e. with particle count), with a steeper increase in portion range τ_{rg} . This allows the interpretation, that in the given setup, the main effect of sample mass increase is elongation of the particle portion. This could be due to the fact, that the horizontal surface available is constant, so the portion is forced to use up more room vertically when there are more particles. With surface area being a limiting factor, we can also expect the average residence time $\overline{t_{res}}$ to increase with sample mass, which is indeed what we see.

The standard deviations between the randomness-prone simulation reruns are shown in Figure 7 in percent relative to the respective goal variable. Relative values for reproducibility are especially important for numerical optimization, because they influence optimizer stability. The values shown correspond to the standard deviation $\sigma_{set,1}$ for a set size of $m = 1$. We find a notable influence of the randomness, with a standard deviation of up to 5.3%. The variability of τ_{rg} is the highest, followed by τ_{sd} and $\overline{t_{res}}$. This goes well together with the expected relationship between variable robustness versus number of particles that influence the goal variable. In the same manner, we obtain less random variations between the runs as sample mass increases. Thus a more reliable result for the respective goal variable is obtained with higher particle counts. An exception to this is the average residence time $\overline{t_{res}}$, whose relative standard deviation slightly increases over sample mass. One possible reason for this could be, that with more particles and thus less space available per particle, the entire portion gets slowed down by friction and particles blocking each other. This short clogging would then take gravity some time to overcome, before the whole portion can continue flowing. The effect however appears very minor to justify further investigation at this point.

3.3 Statistical Evaluation

As discussed above, if we rerun an identical simulation (while implementing the randomness we expect from the real process), we can expect randomness to average out, getting more reliable results. The 300 reruns of every scenario are grouped into variable sets according to Figure 4 and the mean goal variables are calculated for every set of $m = 1 \dots 10$ runs. The standard deviation $\sigma_{set,m}$ between the averaged results of m runs is calculated for every goal variable. Doing so allows assessing which accuracy would be obtained, depending on how often the simulation is reran. The results for the simulations with a sample mass of 500 grams are shown in Figure 6.

Influence of Set Size We see in Figure 8(A) that as expected, the standard deviation $\sigma_{set,m}$ generally decreases with larger set size due to cancelling out of the random changes in the goal variables. We also find that the highest effect of increasing set size manifests for smaller m . So if simulation time is an issue, which in practice it often is, working with a smaller set size maintains good reproducibility. At set size $m = 6$, the instability of τ_{rg} decreased by 42% compared to $m = 1$. For bigger set sizes, variability drops down to around 1%.

We find that the variation in portion range duration τ_{rg} is always greater than the one of standard deviation of the residence τ_{sd} . This supports the prior assumption that the more particles are involved in the calculation of the goal variable, the less sensitive it will be to randomness. $\overline{\tau_{res}}$ is again smaller than τ_{sd} , presumably because the mean is less sensitive to outliers than the standard deviation.

Physical versus Numerical Randomness Figure 8 shows that the effect of the randomness in initial state (A) is greater than the effect of numerical noise (B). We also find that in the scenario with both physical and numerical randomness at play, the standard deviation is almost identical to the case with randomness only coming from the initial state (C). We therefore conclude that numerical noise can be neglected for the process at hand. This is a promising finding, because numerical noise is after all an unphysical artifact, whereas the randomness in initial condition represents a real-life phenomenon and was therefore added to the simulations intentionally.

At larger set size m , we find that $\sigma_{set,m}$ is sometimes greater than the previous $\sigma_{set,m-1}$. This can be explained by the fact that the value for the standard deviation is a mere estimate, as obtaining the true value would require an infinite number of sets of runs. In order to get as good as possible estimates, all available runs from each set are used respectively. This means that sets $\frac{300}{m}$ sets of

runs are taken into consideration. This however reduces the accuracy of the estimate for larger m , since less samples (i.e. sets of runs) are available to calculate $\sigma_{set,m}$.

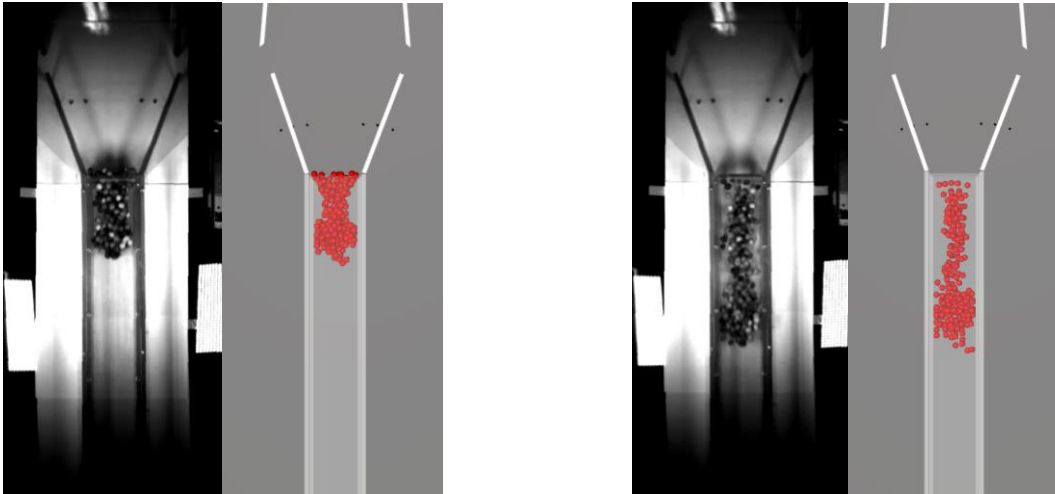


Figure 5: Comparison between experiment and simulation, 250 g after 0.3 s and 0.35 s respectively

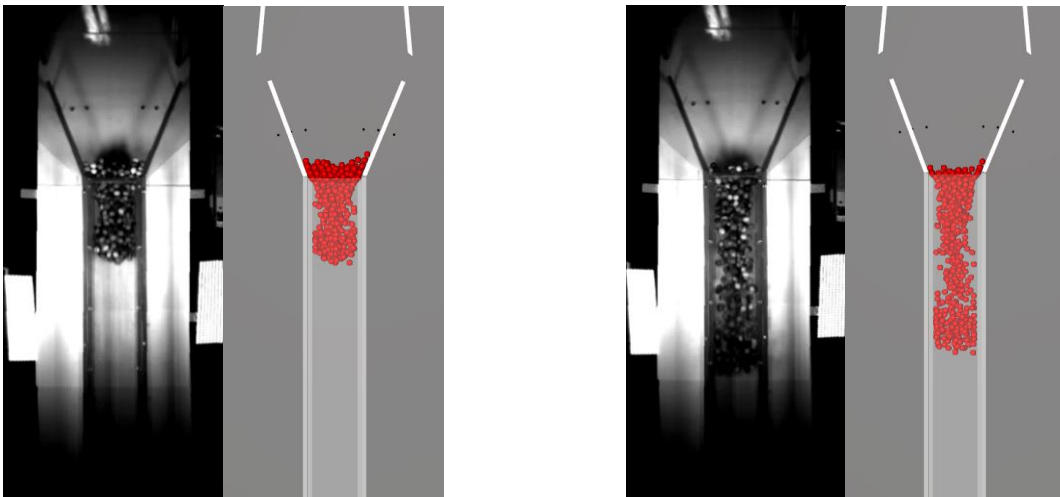


Figure 6: Comparison between experiment and simulation, 500g after 0.3s and 0.35 s respectively

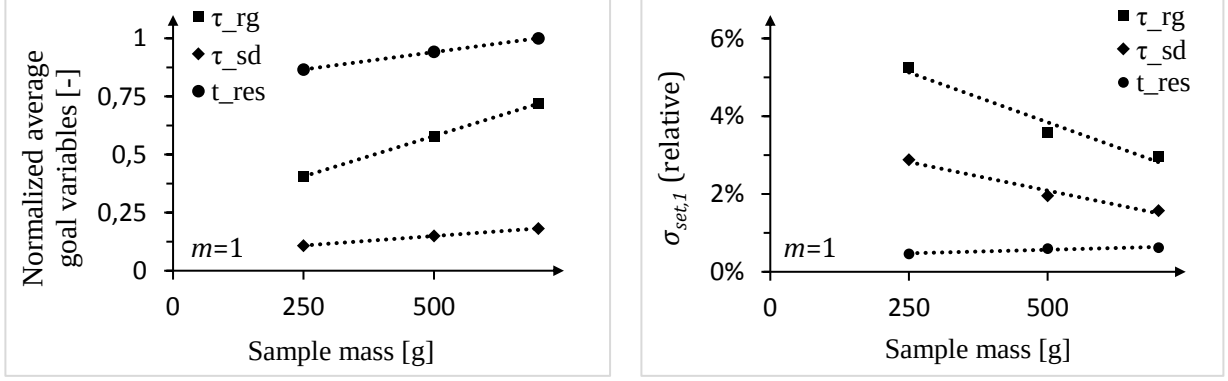


Figure 7: Mean goal variables τ_{rg} , τ_{sd} and $\overline{t_{res}}$ over sample mass and their estimated relative standard deviation $\sigma_{set,1}$ due to randomness in initial state and numerical noise for 300 simulation runs.

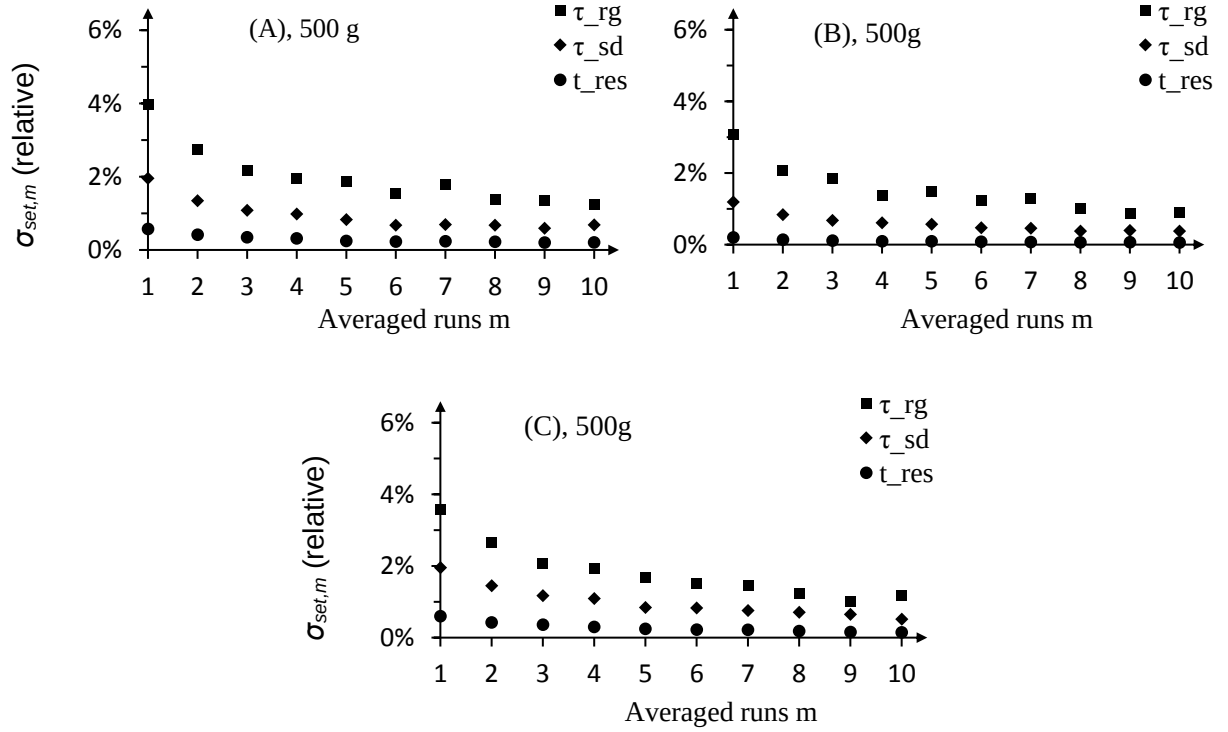


Figure 8: Estimation for the relative standard deviation $\sigma_{set,m}$ due to randomness in initial state (A), numerical noise (B) and both combined (C), based on $\frac{300}{m}$ sets of runs for 500 g

4 Outlook and Summary

4.1 Improvement of Model Accuracy

As long as the model parameters allow physically plausible simulation results, the amount of variability obtained in the process is not considered to be affected by model accuracy. On the other hand, when the findings of this study should be employed in answering practical design questions, high model fidelity is desirable.

It is very challenging to analytically determine parameters for DEM models. Many studies address this issue by iteratively calibrating the model to a real life observation of the bulk material. By doing so, the simulation accuracy can be improved [22, 25, 26]. However in the partially random VFFS filling process, if every iteration was only conducted once, one would run the risk of finding inaccurate parameters due to random variations that do not depend on the actual model parameters. However, as we showed it is possible to reduce the influence of randomness by conducting several simulation runs and averaging the results. For the above discussed goal variables, there is no measurement technique available in literature, but other processes with different ones could be used.

Markauskas et al. presented a calibration technique for a DEM model of rice, which involves tracking the fill level of a transparent hopper filled with the granulate and then being drained through the bottom [27]. Another method frequently found in literature, is the angle of repose test, where the angle of a heap of the bulk material is used as a reference. [28, 29]. Future studies should assess if these or other methods are suitable to determine more accurate model parameters for the process at hand.

4.2 Summary

In this study we set out to evaluate the challenges in DEM simulations of a typical VFFS filling process, not due to shortcomings in the physical model, but due to both physical and numerical randomness. To assess feasibility, we started out by designing a simplified experimental setup representing the real process. We conducted drop tests with bite-size chocolate candy with sample masses between 250 and 700 grams. The experimental setup was reproduced in a DEM computer model. Model parameters were obtained from literature on similar materials, measurements and estimation.

We obtained a good match between simulations and experiments. Through statistical evaluation, we showed that high reproducibility of simulation results can be achieved by averaging several randomness-prone simulation reruns. We found that the randomness in initial conditions outweighed the randomness due to numerical noise. Furthermore, it was shown that even potentially unstable goal variables, depending on only two particles, can be determined with good accuracy, if a sufficient number of reruns is conducted. Result stability is generally higher when working with greater sample masses i.e. particle counts. The results allow for future productive simulation of the process at hand or similar ones, which may assist engineers in design and optimization tasks.

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